DBMS Lab Assesment Report

Placement Portal Application

1. Students Details

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2. Project Details

Project Title: Placement Portal Application

Problem Statement:

To design and build a web application that allows **Admin (Institute), Company, and students** to interact with the system based on their roles.

Approach:

The platform was built using Flask with a modular structure to handle role-based authentication, profile management, and backend data processing. It allows companies to post placement drives, students to submit job applications, and stores all records securely in a structured relational database. Role-specific dashboards (Admin, Company, and Student) help users track application statuses, drive approvals, and new notifications. The system ultimately focuses on streamlining the campus recruitment process, enabling seamless communication and providing centralized tracking for placement activities.

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3. AI/LLM Declaration

I used **ChatGPT (GPT-5)** to assist in writing SQLAlchemy model definitions, creating API documentation samples, and improving variable naming consistency.

The extent of AI/LLM usage is around **15% - 20%,** limited to **code suggestions and documentation formatting**.
All final implementation logic, debugging, and integration were done manually.

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4. Technologies and Frameworks Used

Technology / Library	Purpose in the Placement Portal
Flask	Core backend framework for routing and app logic.
Flask-SQLAlchemy	ORM for database interaction and managing relationships.
SQLite	Local database for storing users, profiles, drives, and applications.
Jinja2	Template engine for rendering dynamic HTML pages.
HTML/CSS & Bootstrap	Responsive UI design and styling.
Flask Session Management	User authentication and session management.

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5. Database Schema / ER Diagram

1. **User** Stores basic user credentials, system roles, and account status.

   Fields: id, username, email, password_hash, role, is_active, created_at

2. **Student** Stores detailed student profiles, academic information, and resumes.

   Fields: id, user_id, full_name, phone, degree, branch, cgpa, skills, resume_path,

   is_blacklisted, created_at

3. **Company** Stores company details, HR contact information, and account approval status.

   Fields: id, user_id, company_name, industry, website, hr_contact_name,

   hr_contact_email, hr_contact_phone, description, approval_status, is_blacklisted,

   created_at

4. **PlacementDrive** Stores the job postings, required skills, and deadlines created by approved companies.

   Fields: id, company_id, job_title, job_description, eligibility_criteria, required_skills, experience_required, salary_range, application_deadline, status, created_at

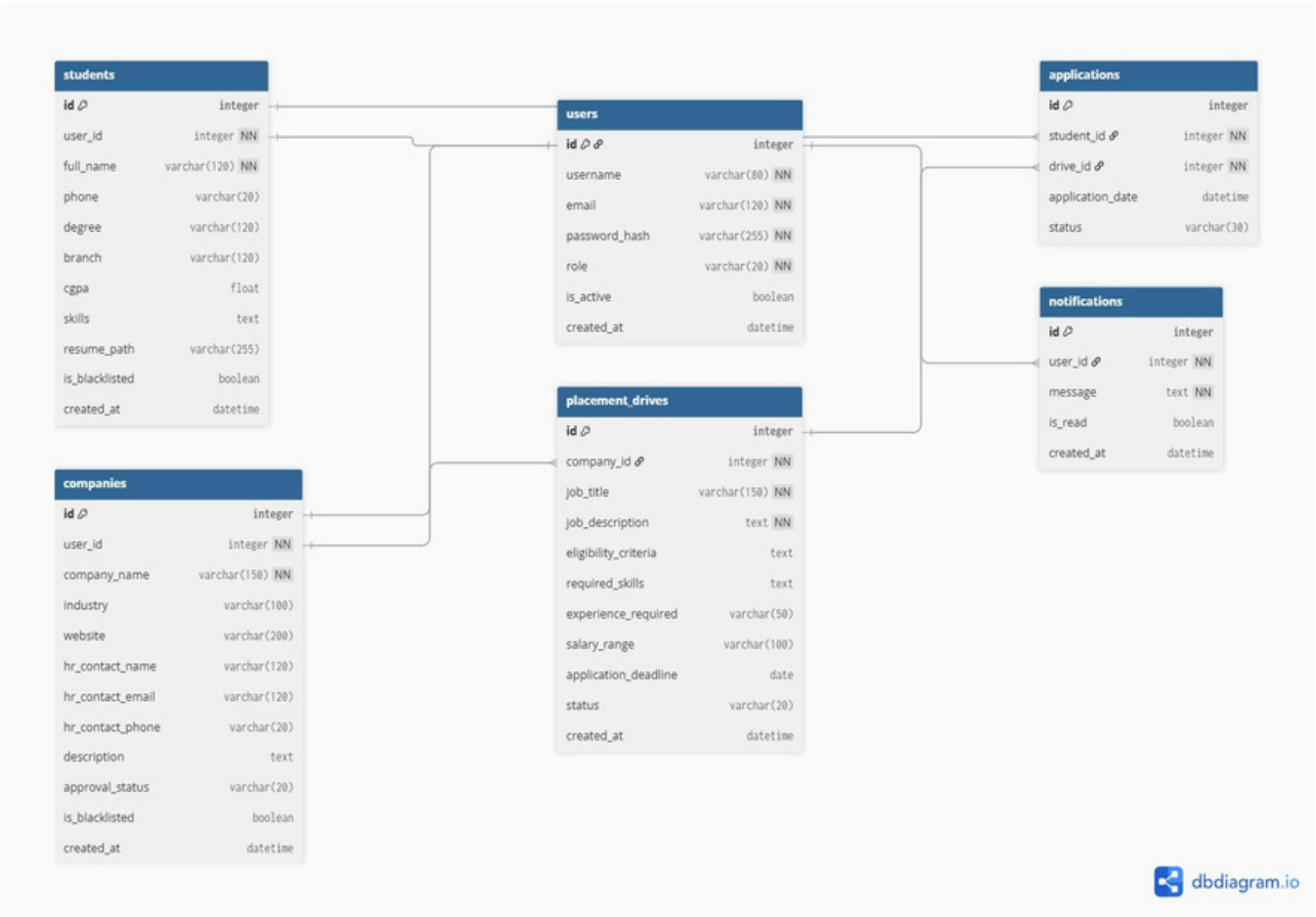
5. **Application** Tracks the applications submitted by students to specific placement drives, including their hiring status.

   Fields: id, student_id, drive_id, application_date, status

6. **Notification** Stores system alerts and messages sent to users (Admins, Companies, or Students).

   Fields: id, user_id, message, is_read, created_at

Relationships: ER diagram



6. API Resource Endpoints

Endpoint	Method	Description
/auth/login	GET, POST	Authenticates users and starts a secure session.
/auth/register/student	GET, POST	Registers a student and handles resume uploads.
/auth/register/company	GET, POST	Registers a company (pending admin approval).
/student/drives/<id>/apply	POST	Submits a student's job application.

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7. Architecture

Architecture Overview:

- app.py – Main entry point; initializes the app, database, and routing.
- /controllers/models.py – Defines database tables and relationships using SQLAlchemy.
- /controllers – Flask Blueprints handling routing, authentication, and core logic for all user roles.
- /templates – Jinja2 HTML files for rendering dynamic web pages (dashboards, forms, listings).
- /static – Stores CSS, JS, and Bootstrap files for frontend styling and layouts.

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8. Video Presentation

Drive Link:

https://drive.google.com/file/d/1vg9uTFC4FppX3an9pAB09ZrdZXt3sfAT/view?usp=sharing

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